

Generating Synthetic Data with Diffusion Models for Writer Identification

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Abstract. Writer identification aims to determine the author of a handwritten sample based on individual writing characteristics. In this work, we investigate the use of diffusion-based handwritten text generation for data augmentation in offline word-level writer identification, which is particularly useful when lacking data. Specifically, we employ DiffusionPen to synthesize writer-conditioned handwriting samples and study their impact on writer identification performance in both contemporary and historical settings. Experiments are conducted on the CVL dataset and the historical BullingerDB dataset. Our results show that synthetic augmentation consistently improves writer identification accuracy on binarized images, yielding gains of 1.2 percentage points on CVL and 1.33 on Bullinger. Furthermore, we observe that grayscale image representations outperform alternative input representations across all experimental settings, for which synthetic data does not improve performance. We additionally demonstrate that synthetic data is particularly effective in low-resource scenarios, where the relative improvement increases as the amount of available real training data decreases.

Keywords: Writer identification · Handwriting Synthesis · Synthetic Data · Diffusion Models

1 Introduction

Handwriting is a distinctive biometrics of human motor behaviour, shaped by factors such as motor learning, cultural background, writing habits, and biomechanical characteristics [3]. As a result, each writer develops an individual writing style that remains observable even at the level of a very small amount of handwriting, e.g., an isolated word [4]. Secondly, handwriting is a valuable biometric modality for applications including forensic document analysis, historical manuscript research, and archival studies [14]. Automated Writer Identification (WI) aims to assign a handwriting sample to one of several known writer identities [13]. Our work focuses on offline WI in a word-level setting, where the goal is to identify the writer from a single handwritten word image without considering the textual content.

In this paper, we investigate whether, and to what extent, Handwritten Text Generation (HTG) can be used to improve the training of WI models through

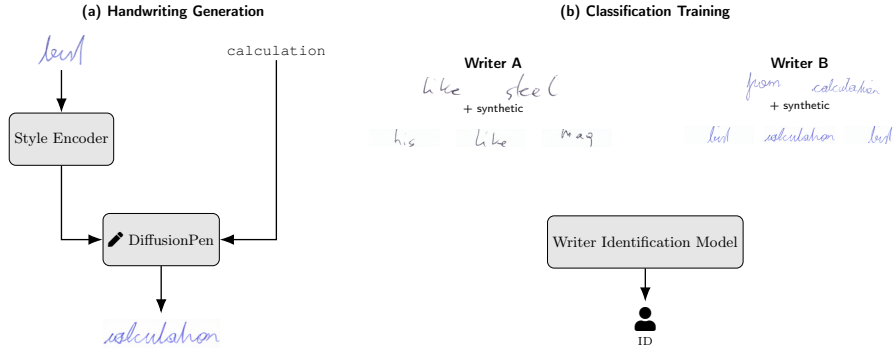


Fig. 1: Overview of our approach. We first generate handwriting with Diffusion-Pen [12] that is style-conditioned by forwarding images of a specific writer. Then, we use the synthetic data to augment the training process of our WI model.

data augmentation. Recent advances in diffusion-based generative models have improved the quality and stylistic consistency of synthesized handwriting [11,12]. In particular, prior work demonstrated that diffusion-generated handwriting preserves writer characteristics significantly better than earlier GAN-based approaches, achieving approximately 70% writer identification accuracy compared to less than 5% for GAN-generated samples [11]. Motivated by these findings, we explore the use of diffusion-based handwriting synthesis to augment the training data for WI across two distinct scenarios: (1) contemporary handwriting using the CVL dataset [6], and (2) historical handwriting using the BullingerDB dataset [14]. An overview of the proposed pipeline is illustrated in Fig. 1. To the best of our knowledge, this is the first study investigating diffusion-based handwriting generation for historical documents in the context of writer identification.

Our experimental results demonstrate that synthetic augmentation improves writer identification performance in both settings, particularly when using binarized images. On the CVL dataset, we observe an improvement of 1.2 percentage points on Top-1 accuracy, while on the BullingerDB dataset we achieve 1.33. Furthermore, our experiments indicate that grayscale representations consistently outperform other image representations for WI, both when training solely on real data and when incorporating synthetic samples. In addition, our findings confirm that synthetic data is particularly beneficial in low-resource scenarios. Similar to observations reported for text recognition [10], we find that the relative benefit of synthetic augmentation increases when the amount of available real training data is reduced. Specifically, when using only 50% of the original training data, synthetic augmentation yields an improvement of approximately 6%.

The rest of our study is structured as follows: Section 2 described related work, Section 3 provides details of our method, and Section 4 shows the results of our approach. We conclude the paper in Section 5.

2 Related Work

Writer Identification WI is the task of assigning a handwriting sample to one of several known writer identities. In the closed-set setting, the writer of the query document is assumed to be present in the database (classification). In the open-set setting, the task is to find similar documents in a given gallery (retrieval). In this work, we focus on WI and present related work in the following.

The first deep-learning-based approach for WI is proposed by Fiel and Sablatnig, in which they use a CNN to identify the writer based on a single line image. The output of the penultimate layer is then used as a feature vector for pooling. This simple method already achieves accuracies of 98.9 % on the CVL [6] dataset.

Further research mostly focuses on word-level identification. He and Schomaker propose FragNet [3], an architecture for word-based WI that is aimed to explore all writing style information contained in the whole word image by adding a second path which takes fragments of the word image as input. Their follow-up work which this paper is based on is Global-Context Residual Recurrent Neural Network (GR-RNN) [4] extends FragNet by adding gated recurrent units. We explain GR-RNN in Section 3.

Koepf et al. [7] investigate the use of vision transformers for the WI task on page-level and find that their application is limited due to the small dataset sizes (CVL [6], ICDAR 2013 [8]).

Recently, WI has been investigated in the CircleID competition at ICDAR2026 [2] for which the writer of a hand-drawn circle has to be determined, with the additional task of recognizing foreign writers as outliers. WI is also closely aligned with the open-set task of *writer retrieval*, for which current research focuses on historical documents [13].

Handwriting Synthesis Generative Adversarial Networks (GANs) have been widely used for offline handwriting generation [1]. Early methods condition generation on text representations and often rely on auxiliary recognisers to improve readability. Later approaches, such as GANwriting [5], jointly condition on text and style references to generate images in seen and unseen writer styles.

Although the handwriting is visually plausible, the style preservation lacks compared to the current dominant architecture of diffusion-based models. The first work is WordStylist, that also applied WI as a evaluation task and show that diffusion models outperform GANs by more than 60% accuracy. The follow-up work DiffusionPen [12] is used for generating the handwriting in this work, and we briefly explain it in our methodology.

Another diffusion-based approach is proposed by Mayr et al. [9] that targets paragraph-level generation.

3 Methodology

Our method includes two parts: First, we train DiffusionPen [12] as shown in Fig. 2 to generate synthetic word images. Then, we apply a WI model and use the

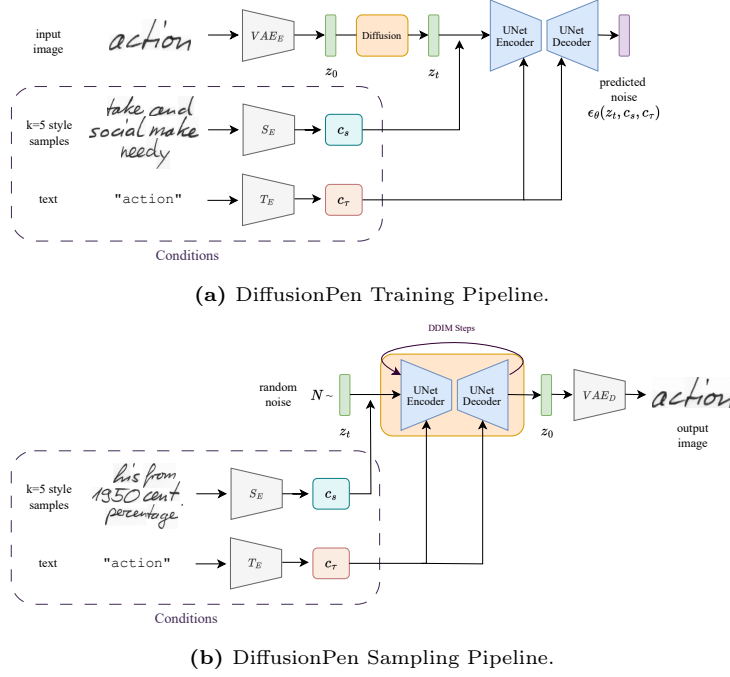


Fig. 2: Overview of DiffusionPen [12]. The model combines a text encoder, a style encoder, and a conditional latent diffusion generator. During training, images are encoded into the latent space; during sampling, generated latents are decoded into handwriting images (Figure taken from [12] via open access version).

synthetic handwriting to augment the training process. We describe the details in the following.

3.1 Diffusion Model

DiffusionPen [12] is a latent diffusion model for few-shot styled handwriting generation. It generates word-level handwriting images conditioned on a text prompt and a small set of reference images from the target writer. The model consists of a latent autoencoder, a character-level text encoder, and a hybrid style encoder. An overview is shown in Fig. 2. The same text encoder as in the original implementation is used (Canine-C). The training process for applying DiffusionPen consists of two stages: style encoder training and denoising model training.

The style encoder is trained to produce compact writer representation from reference images. We use the triplet loss as an objective, which aims to push embeddings from different writers further than embeddings of the same writer.

The text encoder maps the target word into a character-level representation. During generation, the denoising network is conditioned on both the style em-

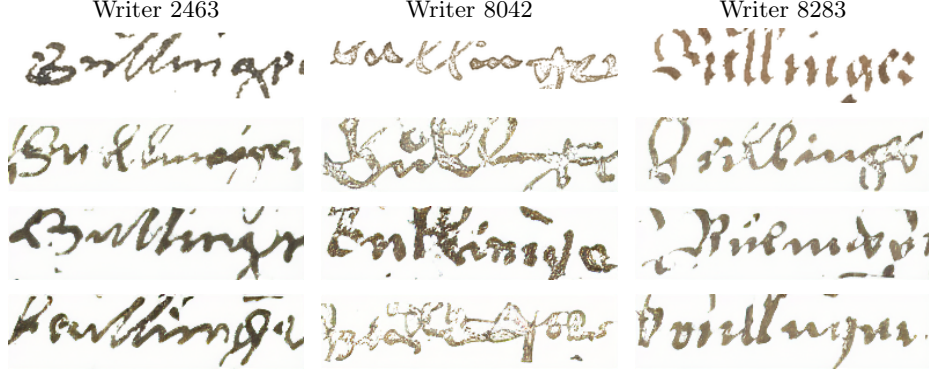


Fig. 3: Real handwriting samples (top row) and generated samples (rows 2–4) for three different writers.

bedding and text embedding. The diffusion model is trained to predict the noise added to a latent representation at timestep t by minimising

$$\|\epsilon - \epsilon_\theta(z_t, c_s, c_\tau)\|^2, \quad (1)$$

where z_t is the noisy latent, c_s is the style condition, and c_τ is the text condition.

3.2 Synthetic Data Generation

After training DiffusionPen, synthetic word images are generated for each writer. For every generated sample, five reference word images from the same writer are randomly selected to define the writer style. The target transcription is sampled from another writer’s word image, excluding transcriptions that already appear for the target writer. This encourages textual diversity while keeping generated words within the vocabulary distribution of the dataset. Examples of the generated samples for the BullingerDB [14] dataset are shown in Fig. 3.

The generated images are then used to train the WI model in two settings: synthetic-only training and mixed training, where synthetic images are added to the real training set. Performance is compared against a real-data-only baseline.

3.3 Writer Identification Model

We adopt the GR-RNN, proposed in [4], as our WI model, as it shows state-of-the-art performance on word-level for contemporary datasets such as CVL [6]. GR-RNN integrates global-context features with local fragment-level information extracted from CNN feature maps to represent handwritten word images. A gated recurrent unit is employed to capture contextual dependencies across the sequence of local fragments, while a residual RNN ensures that each fragment’s representation retains both the between-fragment context and its original

Table 1: Statistics of the datasets used in the experiments.

Dataset	Writers	Total	Train	Test
CVL [6]	310	99,904	79,790	20,114
BullingerDB [14] (Subset)	596	452,640	362,114	90,562

feature. The hidden state of the recurrent network is initialized with a global-context vector obtained via global average pooling of the final convolutional layer. The network takes a single word image as input and classifies the corresponding writer. We use GR-RNN following the original implementation as proposed by the authors.

3.4 Datasets

Experiments are conducted on the CVL [6] dataset consisting of contemporary handwriting and a subset of the historical BullingerDB [14] dataset. The dataset statistics are shown in Table 1.

The CVL dataset contains 310 writers and 99,904 word images. The BullingerDB dataset contains 796 writers and 4,427,128 word images. We construct a smaller subset by removing the 200 writers with the largest number of images, resulting in 596 writers and 452,640 word images (a writer has on average 600 word instances). Note that the BullingerDB corpus is challenging, in particular for generating syntethic data, as the writers contributed over a longer period of time, meaning that the writing style may vary for an individual.

For both datasets, 80% of each writer’s images are used for training and 20% for testing.

4 Evaluation

In the following, we describe our experimental setup and the main results of our paper. While the results are presented on the CVL benchmark, we end the section with a study on the BullingerDB dataset.

4.1 Experimental Setup

All word images are resized to 256×64 pixels. The model is evaluated on the fixed real test set of each dataset. We run experiments with three different seeds and the standard deviation across runs is less than 0.1 percentage points.

Training setup WI model is trained for 50 epochs using the Adam optimizer with a learning rate of 10^{-4} , a weight decay of 10^{-4} , and a mini-batch size of 16. The learning rate is reduced by a factor of 0.5 every 10 epochs.

The style encoder is trained for 100 epochs using Adam with a learning rate of 10^{-3} and weight decay of 10^{-4} . The learning rate is reduced by a factor of 0.1 every three epochs.

The diffusion model is then trained using the frozen style encoder to provide the writer-style condition. We use a DDIM scheduler for noise injection and sampling. During training, the diffusion process uses 500 timesteps. During sampling, the number of reverse steps is reduced to 50. The denoising model is optimised with AdamW using a learning rate of 10^{-4} and weight decay of 0.2.

Data Generation Synthetic data volume is expressed relative to the size of the real training set. For example, 100% generated data means that the number of generated images per writer equals the number of real training images for that writer.

4.2 Style Preservation

First, we check the style preservation of DiffusionPen for the given writers. We take a GR-RNN trained on real CVL data and use it to classify the generated images, i.e., a train-on-real, test-on-synthetic protocol. Table 2 reports the results for both image representations, with the generated set sized at 100% and 200% of the real training data. We notice a strong performance gap between the two formats. We assume that the grayscale model partly relies on ink and pen cues that the generator, rendering in generic ink, does not reproduce, whereas the binarised model relies on stroke-level style, which the synthetic images do preserve. We use this as motivation for the format-dependent behaviour observed throughout the remaining experiments.

Table 2: Style preservation on CVL. GR-RNN trained on real data classifies the generated images.

Image type	Generated ratio	Top-1	Top-5
Grayscale	100%	9.17	22.67
Grayscale	200%	5.03	16.05
Binarised	100%	78.19	91.85
Binarised	200%	78.08	91.68

4.3 Effect of Image Format

We focus on WI performance on the CVL dataset under different configurations. We investigate three factors: the image format used to train DiffusionPen, the input format used for GR-RNN, and the amount of synthetic data added to the real training set. Table 3 reports the Top-1 accuracy for different combinations

Table 3: Top-1 accuracy (in %) on CVL writer identification test set for different DiffusionPen image formats and GR-RNN input formats.

	DiffusionPen	GR-RNN	Syn. only	Real Only	Syn. + Real
RGB	Grayscale	66.76	94.29	94.09	
RGB	Binarised	73.78	82.87	83.98	
Grayscale	Grayscale	49.84	94.29	89.96	
Grayscale	Binarised	50.75	82.87	80.23	
Binarised	Grayscale	59.56	87.17	84.47	
Binarised	Binarised	40.50	82.87	83.65	

of DiffusionPen training and GR-RNN input format. Three training conditions are compared: generated data only, real data only, and real data combined with 100% generated data.

The results show that synthetic data improves performance when added to real data using binarized data. The real-only baseline for binarised GR-RNN is 82.87%, independent of DiffusionPen’s training format, since it does not involve synthetic data. The strongest improvement is observed when DiffusionPen is trained on RGB images and GR-RNN on binarised images, where Top-1 accuracy increases from 82.87% to 83.98%. In contrast, synthetic data generated by the grayscale DiffusionPen model reduces performance in both evaluated GR-RNN input formats, suggesting that not all generated data is equally beneficial. The state-of-the-art performance of GR-RNN is also achieved on grayscale images (92.6% [4]). We believe that networks trained on RGB and grayscale images infer features from ink patterns or even different pens from the word images instead of the handwriting style [13].

4.4 Effect of Synthetic Data Volume

Table 4 reports Top-1 and Top-5 accuracy for RGB-trained DiffusionPen under different synthetic data volumes. Adding synthetic data slightly improve Top-5 accuracy for grayscale images, while we also observe that for binarized word images, the mixed scenario outperforms the networks trained on real data only. Using only synthetic data, the networks trail by approximately 28% (grayscale) and 10% (binarized). Fig. 4 additionally tracks Top-1 accuracy as the amount of generated data is raised from 10% to 500% of the real training set using binarised images. Both conditions improve up to around 200% and then decline, with the augmented condition eventually falling back below the real-only baseline. Beyond a moderate multiple of the real data, additional synthetic samples begin to dilute rather than reinforce the real signal, so there is a sweet spot rather than a monotone gain.

Table 4: Effect of synthetic data volume on CVL writer identification accuracy (in %) using RGB-trained DiffusionPen for the GR-RNN model.

Training Data	Grayscale		Binarised	
	Top-1	Top-5	Top-1	Top-5
Real Only	94.29	98.66	82.87	93.98
100% Syn. + Real	94.09	98.70	83.98	94.65
200% Syn. + Real	93.97	98.67	84.07	94.97
100% Syn. Only	66.76	84.37	73.78	89.08
200% Syn. Only	66.57	84.43	78.99	92.12

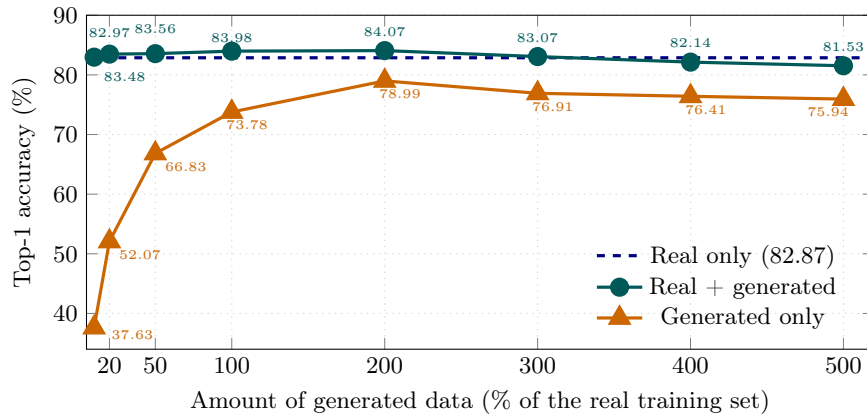


Fig. 4: Top-1 accuracy on the CVL test set (binarised) as the amount of generated data grows from 10% to 500% of the real training set.

4.5 Effect of Real Data Availability

Lastly, we show the effect of having less real data available during training. Fig. 5 illustrates the effect of reducing the amount of real training data used to train the diffusion and WI model while keeping the generated data volume fixed. Firstly, we observe that the mixed scenario always outperforms the real only baseline except for 60%, the gap is the biggest for 20% of the training data (28% accuracy) and smallest for 100% (0.1%). In low data scenarios, even the generated only baseline performs better. We therefore conclude that the synthetic data can successfully replace the real data, or can even be added to improve the WI performance.

4.6 Historical Handwriting Generation: BullingerDB

We also provide baseline experiments for WI on the BullingerDB [14], shown in Table 5. We observe that the margins of mixed and real only are smaller (2-3% accuracy), and similarly to contemporary handwriting, adding synthetic data

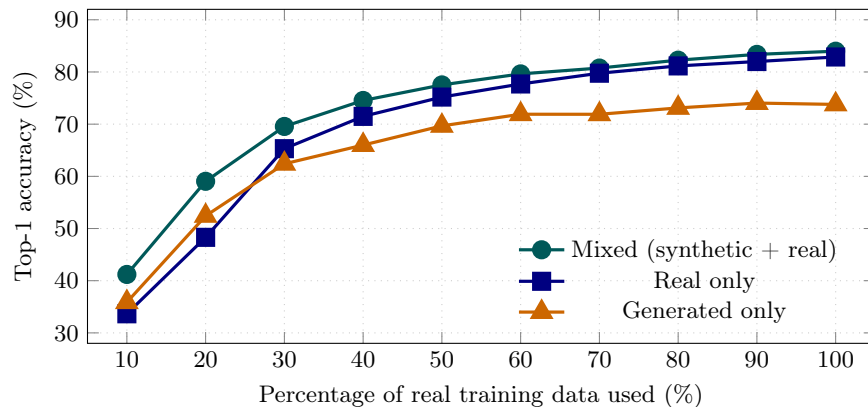


Fig. 5: Top-1 accuracy (binarised) on CVL as the fraction of real training data varies. The volume of generated data is fixed at 100% of the full training size.

Table 5: Top-1 accuracy (in %) on BullingerDB WI for different image formats.

	GR-RNN Syn. only	Real Only	Syn. + Real
Grayscale	25.72	88.04	85.80
Binarised	44.83	77.29	78.62

improves performance for binarized images, making our diffusion-based approach particularly attractive for identifying writers purely on stroke patterns instead of ink or pen characteristics.

5 Conclusion

This paper investigated whether diffusion-generated handwriting images can be used as training data for writer identification. The results show that synthetic word images can improve writer identification accuracy when added to real data, but the benefit depends on the image format used during generation and identification.

For the CVL dataset, synthetic data generated by an RGB-trained Diffusion-Pen model consistently improves GR-RNN performance when combined with real data for binarized data. Increasing the volume of synthetic data further improves both Top-1 and Top-5 accuracy. In low data scenarios, adding synthetic data also helps improve the performance. Our results on BullingerDB showed that diffusion-based HTG is a viable alternative, where the results are similarly to the CVL dataset.

Our future work will study additional evaluation for synthetic data (e.g., style interpolation to generate new writers), and a broader investigation of diffusion-based models for HTG applied to WI.

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